

ScentML: A Machine Learning Engine for Molecular Olfaction Prediction and Fragrance Ingredient Recommendation

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Abstract—We present ScentML, a multi-label machine learning system for predicting perceived odor profiles of chemical compounds from their molecular structure. Trained on 3,755 expert-labeled molecules from the GoodScents olfactory database, ScentML uses Morgan fingerprints (ECFP4, 2048-bit) as molecular representations and a MultiOutputClassifier of gradient-boosted trees (XGBoost) to predict 154 binary odor descriptors per molecule. The system achieves a mean AUC of 0.826 across all labels. We additionally introduce a reverse ingredient recommendation module that retrieves molecules matching a target scent profile via cosine similarity over predicted odor vectors. An interactive Streamlit application makes both capabilities accessible to non-technical users. ScentML demonstrates the viability of data-driven methods for fragrance ingredient discovery, a domain historically dependent on expert human perfumers.

Index Terms—olfaction, molecular fingerprints, multi-label classification, XGBoost, fragrance, SMILES, cheminformatics, SHAP

I. INTRODUCTION

Olfaction is one of the least computationally addressed human senses. Unlike vision and language, where deep learning now achieves human-level performance, the prediction of smell from molecular structure remains an open research problem [1]. The global fragrance industry, valued at over \$50 billion, continues to rely primarily on expert perfumers who develop intuition for structure-odor relationships over decades of experience.

Recent advances in molecular representation learning and multi-label classification have created an opportunity for systematic, data-driven olfaction modeling. Several studies have demonstrated that Morgan fingerprints and graph-based representations can capture structure-odor relationships at scale [2], [3]. However, most prior work focuses solely on forward prediction (structure $\rightarrow$ odor) and lacks a corresponding reverse formulation useful for practical ingredient discovery.

ScentML addresses both directions: given a molecular structure, predict its odor profile; and given a target odor profile, recommend the best matching molecules from a curated database. This bidirectional design mirrors the two core workflows of a fragrance R&D chemist.

II. RELATED WORK

Keller et al. [1] established a benchmark for olfactory prediction using the DREAM Olfaction Challenge dataset,

demonstrating that ensemble methods over molecular descriptors can approach inter-human consistency. Sanchez-Lengeling et al. [2] applied graph neural networks to the same task, showing that learned molecular representations outperform hand-crafted fingerprints on small datasets. Lee et al. [3] introduced contrastive learning for odor representation, achieving state-of-the-art results on the GoodScents dataset.

ScentML differs from these works in its practical orientation: rather than maximizing benchmark performance, it prioritizes an end-to-end system, from data preprocessing through deployment, with an interpretable model and a bidirectional user interface accessible without chemical domain expertise.

III. DATASET

A. Source

Data was sourced from the GoodScents Company olfactory database, accessed via the PyrFume open-science Python library [4]. GoodScents contains odor descriptors assigned by professional flavorists and perfumers for thousands of commercially relevant compounds.

B. Identifier Bridging

A key preprocessing challenge was the mismatch between identifier systems: GoodScents behavioral labels use CAS Registry Numbers, while molecular structure records use PubChem Compound IDs (CIDs). We resolved this using PubChem’s batch synonyms REST API to map CIDs to CAS numbers, enabling cross-table joins without additional packages.

C. Final Dataset Statistics

After filtering for molecules with both structural data and at least one odor label, and removing descriptors with fewer than 15 positive examples:

- **Molecules:** 3,755
- **Odor descriptors:** 154
- **Train / Test split:** 80% / 20% (stratified)
- **Feature dimensionality:** 2,048 (Morgan fingerprint bits)

IV. METHODOLOGY

A. Molecular Representation

Each molecule is encoded as a 2,048-bit Morgan fingerprint with radius $r = 2$, equivalent to ECFP4 [5]. This circular

fingerprint encodes the local chemical environment of each atom up to two bond hops, producing a fixed-length binary vector. Generation was performed using RDKit’s `rdFingerprintGenerator.GetMorganGenerator` API.

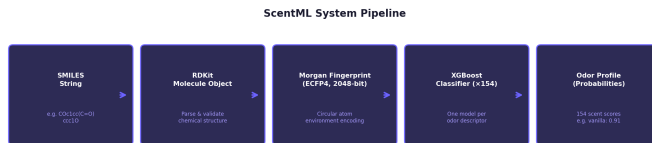


Fig. 1. ScentML system pipeline. A SMILES string is converted to a 2,048-bit Morgan fingerprint, passed through a multi-output XGBoost classifier, and mapped to 154 odor descriptor probabilities.

B. Multi-Label Classification

We formulate odor prediction as a multi-label binary classification problem with 154 independent targets. A `MultiOutputClassifier` wrapper from scikit-learn trains one XGBoost binary classifier per odor descriptor. XGBoost was selected for:

- Strong performance on sparse binary tabular features
- Built-in L_1/L_2 regularization to mitigate overfitting on rare labels
- Compatibility with SHAP TreeExplainer for post-hoc interpretability

C. Decision Threshold Tuning

The default classification threshold of 0.5 is suboptimal for imbalanced multi-label data. We tuned the threshold τ over the range $[0.1, 0.9]$ on the validation set, selecting $\tau^* = 0.30$ to maximize macro-averaged F1 score across all 154 labels.

D. Reverse Ingredient Recommendation

Each molecule m in the database is represented as a 154-dimensional odor probability vector $\mathbf{p}_m \in [0, 1]^{154}$, obtained from model predictions. Given a user-specified target odor vector $\mathbf{q} \in [0, 1]^{154}$ (constructed from selected descriptors and intensity weights), candidate molecules are ranked by cosine similarity:

$$\text{score}(m) = \frac{\mathbf{q} \cdot \mathbf{p}_m}{\|\mathbf{q}\| \cdot \|\mathbf{p}_m\|} \quad (1)$$

The top- N molecules are returned as ingredient recommendations.

E. Explainability

SHAP (SHapley Additive exPlanations) TreeExplainer [6] was applied to decompose individual predictions into per-feature contributions. This allows identification of which molecular substructures (fingerprint bits) most strongly drive a given odor classification, providing scientifically interpretable outputs for domain experts.

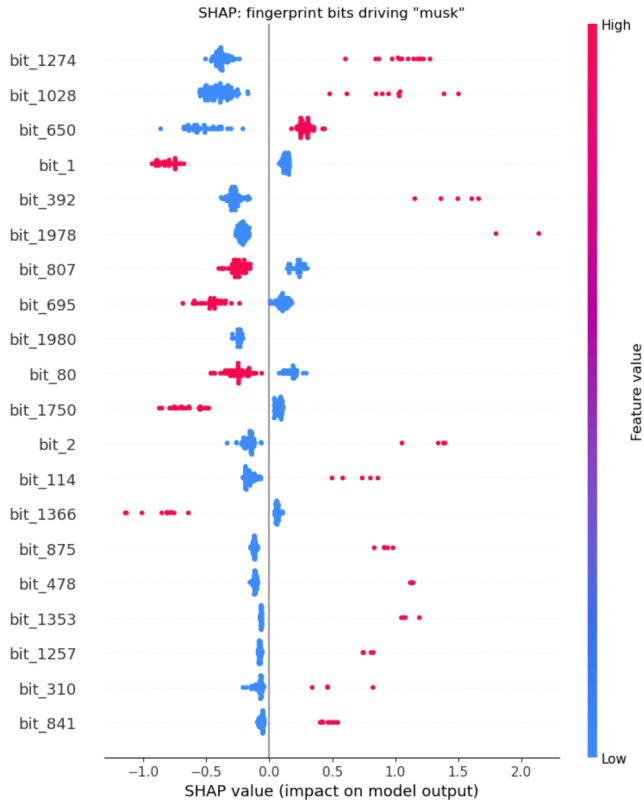


Fig. 2. SHAP feature importance for the “musk” odor descriptor. Red points indicate molecules where the fingerprint bit is present (high feature value); blue indicates absent. Points to the right of zero increase the predicted musk probability.

V. RESULTS

A. Classification Performance

TABLE I
MODEL EVALUATION METRICS (TEST SET)

Metric	Value
Mean AUC (154 labels)	0.826
F1 Micro ($\tau = 0.30$)	0.378
F1 Macro ($\tau = 0.30$)	0.216

A mean AUC of 0.826 indicates the model reliably ranks true odor labels above false ones across descriptors. The moderate F1 scores reflect inherent challenges: severe class imbalance (most descriptors have $<5\%$ positive rate), label noise from subjective human annotation, and the intrinsic difficulty of the structure-odor mapping problem.

B. Qualitative Validation

Predictions on well-known fragrance molecules are chemically consistent:

- **Vanillin** (vanilla): top predicted labels: *vanilla*, *sweet*, *phenolic*, *caramellic*
- **Linalool** (lavender): top predicted labels: *floral*, *fresh*, *woody*, *citrus*

- **Limonene** (citrus peel): top predicted labels: *citrus, fresh, orange, green*
- **Eugenol** (clove): top predicted labels: *spicy, clove, phenolic, woody*

VI. APPLICATION

ScentML is deployed as a Streamlit web application with two interactive modules:

Forward Predictor: Users input a SMILES string (or select from curated examples including vanillin, linalool, geraniol, limonene, eugenol, and menthol) and receive: (1) a rendered molecular structure, (2) predicted odor descriptor tags above a user-adjustable sensitivity threshold, and (3) a ranked probability bar chart across all 154 descriptors.

Reverse Recommender: Users select target odor descriptors and set intensity weights. The system returns the top matching molecules with match scores, a downloadable results table, and a molecular structure grid. Six preset fragrance accords are provided (Woody Musk, Fresh Citrus Floral, Rose Floral, Oriental Amber, Aquatic Marine, Warm Gourmand) for rapid exploration.

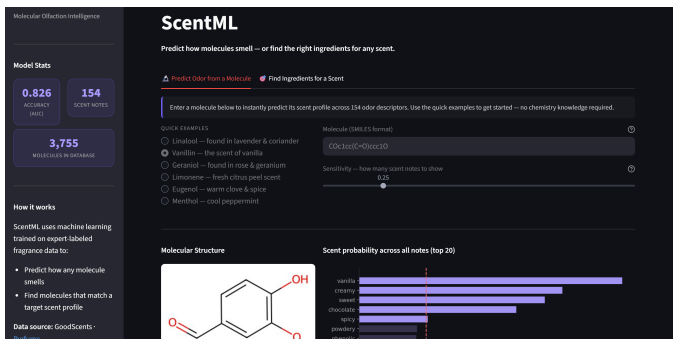


Fig. 3. ScentML Streamlit interface showing the forward prediction tab. The molecular structure of vanillin is rendered alongside predicted scent descriptor probabilities across 154 odor notes.

VII. LIMITATIONS AND FUTURE WORK

The current system has several limitations. Morgan fingerprints encode 2D molecular topology but do not capture 3D conformational geometry, which is known to influence olfactory receptor binding. Label noise from subjective human annotations reduces training signal quality. The dataset covers single-molecule odors; real perfumery involves complex mixtures with emergent olfactory properties not modeled here.

Future directions include: (1) integration of 3D molecular descriptors or graph neural networks (GNNs) for richer representations; (2) training on larger datasets incorporating proprietary industry data; (3) mixture odor modeling; (4) contrastive pretraining on unlabeled molecular data; and (5) a natural-language conversational interface for fragrance design.

VIII. CONCLUSION

ScentML demonstrates that machine learning can model the structure-odor relationship at scale with competitive AUC and

meaningful qualitative predictions. Its bidirectional architecture, predicting odor from structure and recommending structures from odor, directly addresses the two core workflows of fragrance R&D. The accompanying interactive application makes these capabilities accessible to both technical and non-technical users. We believe this work represents a practical step toward data-driven ingredient discovery in the fragrance industry.

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